ECE 2305 – Introduction to C Programming
Tutorial 09
Boolean Operations

Write a C++ application to apply the following Boolean functions and display the results.

```
int a = 10;
int b;  // the variable b will take the values 5, 10 and 15
bool q;

q = a < b;
q = a <= b;
q = a == b;
q = a >= b;
q = a > b;

q = b > 5 && b > 15;
q = b > 5 && b < 15;
q = b < 5 && b > 15;
q = b < 5 && b < 15;

q = b > 5 || b > 15;
q = b > 5 || b < 15;
q = b < 5 || b > 15;
q = b < 5 || b < 15;
```

Write the program to display the value of **q**. Compare the results for **b = 5**, **b = 10** and **b = 15**.

Input

```
// ECE 2305 - Tutorial 09
#include <iostream>
using namespace std;
int main() {
    int a = 10;
    int b;

    cout << "a = " << endl;
    cout << a << endl;

    b = 5;
    cout << "b = " << endl;
    cout << b << endl;
    cout << "a < b: " << (a < b) << endl;
    cout << "a <= b: " << (a <= b) << endl;
    cout << "a == b: " << (a == b) << endl;
    cout << "a >= b: " << (a >= b) << endl;
    cout << "a > b: " << (a > b) << endl;

    cout << "b > 5 && b > 15: " << (b > 5 && b > 15) << endl;
    cout << "b > 5 && b < 15: " << (b > 5 && b < 15) << endl;
    cout << "b < 5 && b > 15: " << (b < 5 && b > 15) << endl;
    cout << "b < 5 && b < 15: " << (b < 5 && b < 15) << endl;
    cout << "b > 5 || b > 15: " << (b > 5 || b > 15) << endl;
    cout << "b > 5 || b < 15: " << (b > 5 || b < 15) << endl;
    cout << "b < 5 || b > 15: " << (b < 5 || b > 15) << endl;
    cout << "b < 5 || b < 15: " << (b < 5 || b < 15) << endl;
```

```
    cout << endl;

    b = 10;
    cout << "b = " << endl;
    cout << b << endl;
    cout << "a < b: " << (a < b) << endl;
    cout << "a <= b: " << (a <= b) << endl;
    cout << "a == b: " << (a == b) << endl;
    cout << "a >= b: " << (a >= b) << endl;
    cout << "a > b: " << (a > b) << endl;

    cout << "b > 5 && b > 15: " << (b > 5 && b > 15) << endl;
    cout << "b > 5 && b < 15: " << (b > 5 && b < 15) << endl;
    cout << "b < 5 && b > 15: " << (b < 5 && b > 15) << endl;
    cout << "b < 5 && b < 15: " << (b < 5 && b < 15) << endl;
    cout << "b > 5 || b > 15: " << (b > 5 || b > 15) << endl;
    cout << "b > 5 || b < 15: " << (b > 5 || b < 15) << endl;
    cout << "b < 5 || b > 15: " << (b < 5 || b > 15) << endl;
    cout << "b < 5 || b < 15: " << (b < 5 || b < 15) << endl;

    cout << endl;

    b = 15;
    cout << "b = " << endl;
    cout << b << endl;
    cout << "a < b: " << (a < b) << endl;
    cout << "a <= b: " << (a <= b) << endl;
    cout << "a == b: " << (a == b) << endl;
    cout << "a >= b: " << (a >= b) << endl;
    cout << "a > b: " << (a > b) << endl;

    cout << "b > 5 && b > 15: " << (b > 5 && b > 15) << endl;
    cout << "b > 5 && b < 15: " << (b > 5 && b < 15) << endl;
    cout << "b < 5 && b > 15: " << (b < 5 && b > 15) << endl;
    cout << "b < 5 && b < 15: " << (b < 5 && b < 15) << endl;
    cout << "b > 5 || b > 15: " << (b > 5 || b > 15) << endl;
    cout << "b > 5 || b < 15: " << (b > 5 || b < 15) << endl;
    cout << "b < 5 || b > 15: " << (b < 5 || b > 15) << endl;
    cout << "b < 5 || b < 15: " << (b < 5 || b < 15) << endl;

    cout << endl;

    system("pause");
    return 0;
}
```

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Output

